SET 19: SURVEY RESPONSES ANALYSIS

Dataset	Survey Responses
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	B
2	B	C	A
3	C	A	D

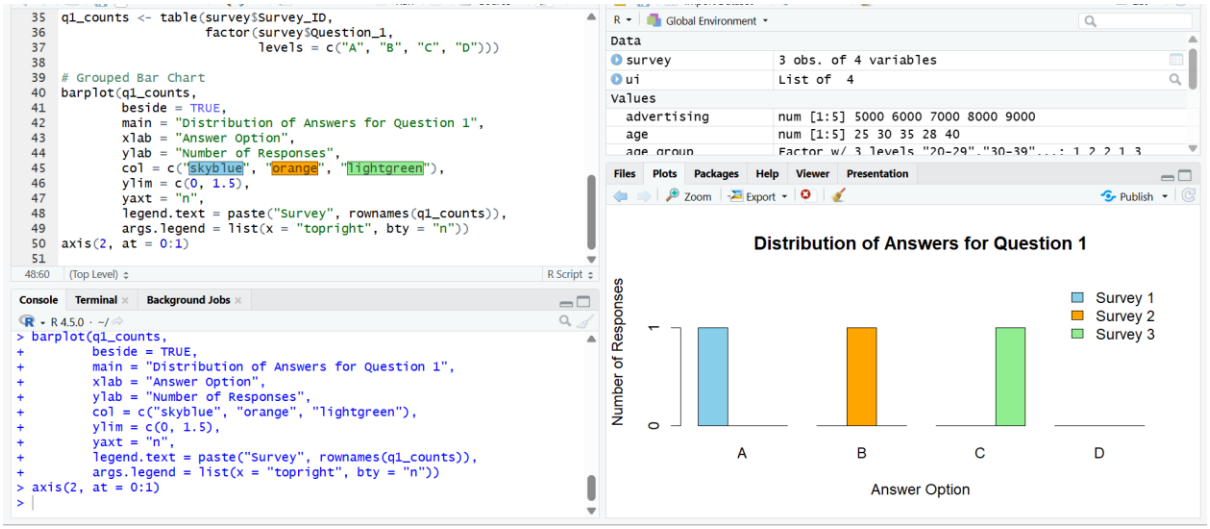
PROGRAM 1: Grouped Bar Chart of Question 1 Answers

Question: Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
# Dataset
survey <- data.frame(
  Survey_ID  = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "C", "A"),
  Question_3 = c("B", "A", "D")
)

# Number of times each answer was chosen in each survey (Question 1)
q1_counts <- table(survey$Survey_ID,
                   factor(survey$Question_1,
                          levels = c("A", "B", "C", "D")))

# Grouped Bar Chart
barplot(q1_counts,
        beside = TRUE,
        main = "Distribution of Answers for Question 1",
        xlab = "Answer Option",
        ylab = "Number of Responses",
        col = c("skyblue", "orange", "lightgreen"),
        ylim = c(0, 1.5),
        yaxt = "n",
        legend.text = paste("Survey", rownames(q1_counts)),
        args.legend = list(x = "topright", bty = "n"))
axis(2, at = 0:1)
```



Output 1: Grouped Bar Chart of Question 1 Answers

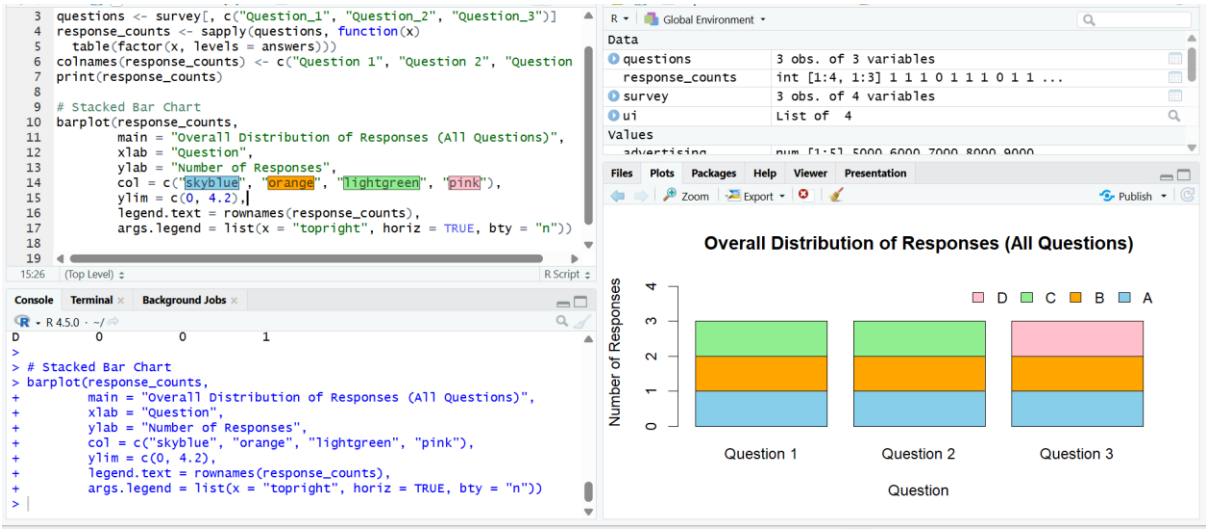
Observation: The grouped bar chart shows the answer chosen for Question 1 in each survey. Survey 1 chose A, Survey 2 chose B and Survey 3 chose C, so each of A, B and C was selected once and D was not selected.

PROGRAM 2: Stacked Bar Chart of All Responses

Question: Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
# Count of each answer option for every question
answers <- c("A", "B", "C", "D")
questions <- survey[, c("Question_1", "Question_2", "Question_3")]
response_counts <- sapply(questions, function(x)
                          table(factor(x, levels = answers)))
colnames(response_counts) <- c("Question 1", "Question 2", "Question 3")
print(response_counts)

# Stacked Bar Chart
barplot(response_counts,
        main = "Overall Distribution of Responses (All Questions)",
        xlab = "Question",
        ylab = "Number of Responses",
        col = c("skyblue", "orange", "lightgreen", "pink"),
        ylim = c(0, 4.2),
        legend.text = rownames(response_counts),
        args.legend = list(x = "topright", horiz = TRUE, bty = "n"))
```



Output 2: Stacked Bar Chart of Overall Responses

Observation: Each question received three responses. Questions 1 and 2 have one response each for A, B and C, while Question 3 has one each for A, B and D. Overall, A and B were chosen three times each, C twice and D once.

PROGRAM 3: Survey Response Data Table

Question: Build a table to show the survey response data for each survey. Label the table elements.

```
# Survey Response Dataset
survey_data <- data.frame(
  Survey_ID  = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "C", "A"),
  Question_3 = c("B", "A", "D")
)

# Display the table
print(survey_data)
```

```r
# Survey Response Dataset
survey_data <- data.frame(
  Survey_ID  = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "C", "A"),
  Question_3 = c("B", "A", "D")
)

# Display the table
print(survey_data)
```

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Console Terminal Background Jobs

R 4.5.0 · ~/

```
+   Question_1 = c("A", "B", "C"),
+   Question_2 = c("B", "C", "A"),
+   Question_3 = c("B", "A", "D")
+ )
>
> # Display the table
> print(survey_data)
  Survey_ID Question_1 Question_2 Question_3
1         1          A          B          B
2         2          B          C          A
3         3          C          A          D
>
```

Output

3: Survey Response Data Table

Observation: The table displays the Survey ID and the answers given to Question 1, Question 2 and Question 3 in each of the three surveys.

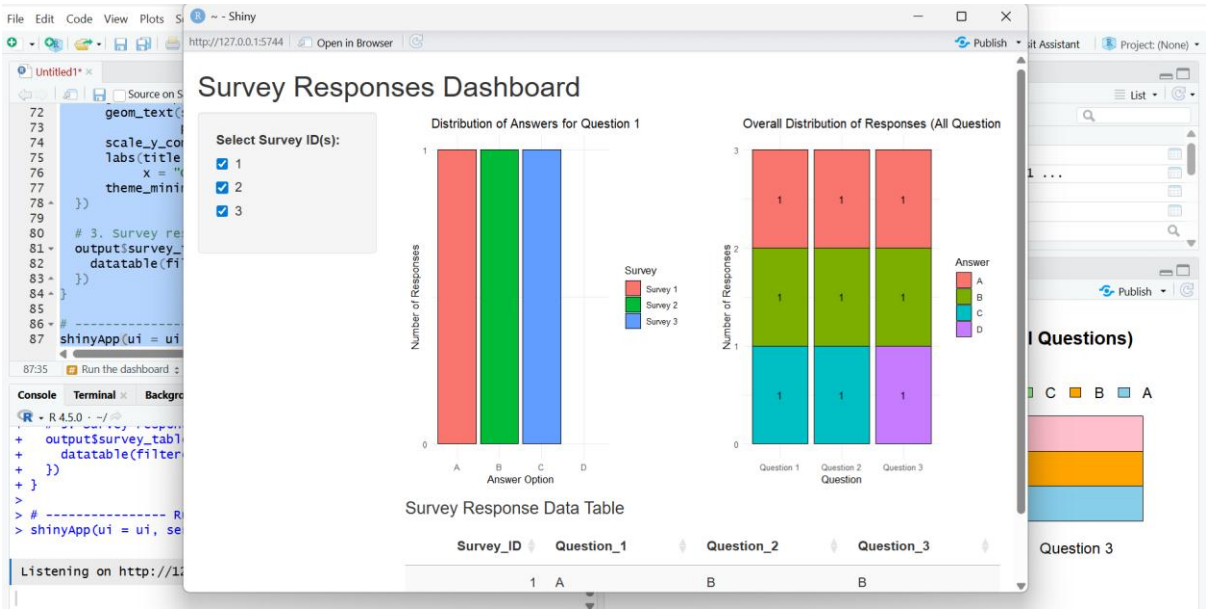
PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Tableau Steps

1. Open Tableau and connect to the Survey Responses dataset.
2. Verify the fields Survey ID, Question 1, Question 2 and Question 3.
3. Select the three question columns, right-click and choose Pivot, then rename the new fields to Question and Answer.
4. Create a worksheet for the grouped bar chart: filter Question to “Question 1”, drag Answer to Columns, Number of Records (or COUNT(Survey ID)) to Rows and Survey ID to Color; select Bar in the Marks card.
5. Add the title “Distribution of Answers for Question 1” and label the axes.

6. Create a second worksheet: drag Question to Columns, Number of Records to Rows and Answer to Color to form the stacked bar chart.
7. Add the title “Overall Distribution of Responses” and label the axes.
8. Create a third worksheet: drag Survey ID to Rows, Question to Columns and Answer to Text to form the survey response table.
9. Create a new Dashboard and add the grouped bar chart, stacked bar chart and table.
10. Add the title “Survey Responses Dashboard”.
11. Use Answer or Question as a filter (click “Use as Filter”) to make the dashboard interactive.
12. Test the dashboard by selecting an answer/question and checking the linked views.



Output 4: Survey Responses Dashboard

Observation: The Tableau dashboard combines the grouped bar chart, stacked bar chart and survey response table, allowing interactive exploration of the survey responses.

Procedure

1. Create the survey response data in R.
2. Count the Question 1 answers for each survey and create the grouped bar chart using barplot() with beside = TRUE.
3. Count the answers for all three questions using sapply() and table(), and create the stacked bar chart using barplot().
4. Create a data frame containing Survey ID and the three question responses and display it as a table.
5. Open Tableau and import the Survey Responses dataset.
6. Pivot the question columns and create the grouped bar chart and stacked bar chart in Tableau.
7. Create the survey response table in Tableau.
8. Combine the three visualizations into an interactive dashboard and add a suitable title.
9. Test the dashboard using the available filter/interactivity.

Result

The Survey Responses data was successfully visualized using a grouped bar chart, stacked bar chart, survey response table, and an interactive Tableau dashboard.